MIL-R-55241A(EL)
11 SEPTEMBER 1964

SUPERSEDES
MIL-R-55241(EL)
27 SEPTEMBER 1963

MILITARY SPECIFICATION

RADIO SET AN/ARC-73( )
UNITS OF

1. SCOPE

1.1 This document covers one type of airborne radio set used for voice communication consisting of Transmitter, Radio T-879 ( )/ARC-73, Receiver, Radio R-1123( )/ ARC-73, Control C-4074( )/ARC-73, and Mounting MT-2699( ) ARC-73, designated as Radio Set AN/ARC-73( ). (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect of invitation for bids, or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111 —Tape, Paper, Gummed (Sealing and Securing).

PPP-B-20 — Bag, Cotton, Mailing.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiberboard.

PPP-B-640 — Boxes, Folding, Fiberboard, Corrugated, Triple Wall.

PPP-F-320 — Fiberboard, Sheet, Stock and Cut Shapes.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).

MILITARY

MIL-E-1 — Electron Tubes and Crystal Rectifiers.

MIL-A-101 — Adhesive, Water Resistant, for Sealing Fiberboard Boxes.

MIL-P-116 — Preservation, Method of.

MIL-B-117 — Bags, Interior Packaging.

MIL-T-713 — Twine and Tape, Lacing and Tying (For Use in Electrical and Electronic Equipment).

MIL-D-3464 — Dessicants, Activated (In Bags) for Static Dehumidification.

MIL-T-5422 —Testing Environmental, Aircraft Electronic Equipment.

MIL-I-6181 — Interference Control Requirements, Aircraft Equipment.

MIL-M-18231 — Marking of Electronic Items.

MIL-F-14072 — Finishes for Ground Signal Equipment.

FSC 5821

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MIL-S-19500 — Semiconductor Devices.

STANDARDS

FEDERAL

No. 595 — Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defect.

MS-20003 — Indicator, Humidity Cord, Three Spot Impregnated Areas (Cobaltous Chloride).

DRAWINGS

SIGNAL CORPS

SC-GL-323019 — Drawing and Gage List for Receiver R-1123( )/ARC-73.

SC-GL-323020 — Drawing and Gage List for Receiver R-1123( )/ARC-73.

SC-GL-323021 — Drawing and Gage List for Transmitter T-879( )/ARC-73.

SC-GL-323022 — Drawing and Gage List for Transmitter T-879( )/ARC-73.

SC-GL-323057 — Drawing and Gage List for AN/ARC-73( ).

SC-DL-451950 — Control, Radio Set C-4074( )/ARC-73.

SC-DL-452149 — Transmitter, Radio T-879( )/ARC-73.

SM-D-452262 — Power Supply Module A4 (Transmitter).

SC-DL-452980 — Receiver, Radio R-1123 ( )/ARC-73.

SM-D-453349 — Power Supply Module A3 (Receiver).

SC-DL-453400 — Mounting MT-2699( ) /ARC-73.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 **Description.** Radio Set AN/ARC-73( ) is an airborne radio communications set which provides transmission and reception of amplitude modulated very high frequency (VHF) radio signals between air to ground and air to air within the range of 108.00 to 151.95 megacycles. Transmission and reception is accomplished by utilization of the following units:

3.1.1 *Transmitter T-879( )/ARC-73.* The transmitter provides the VHF radio signals for transmission in the frequency range of 116.00 to 149.95 mc. Power for transmitter operation is from an integral transistorized power supply that operates from a 28 V power source and supplies DC voltages to the electron tubes. An auto-selector is used for tuning through the frequency range in 50-kilocycle (kc) increments to any one of 680 available channels.

3.1.2 *Receiver R-1123( )/ARC-73.* This unit receives VHF signals within the frequency range of 108.00 to 151.95 mc, amplifies and detects the signals to obtain audio signals. A receiver auto-selector tunes the receiver in 50 kc increments to any one of 880 available channels. The receiver operates from a power supply similar to the transmitter supply.

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3.1.3 *Control, Radio Set C-4074( )/ARC-73.* This unit controls power On-Off, frequency selection, volume control, and squelch control for the transmitter and receiver. (3.1.1 and 3.1.2)

3.1.4 *Mounting MT-2699( )/ARC-73.* The mounting provides support, interconnection. and shock isolation for the transmitter and receiver. (3.1.1 and 3.1.2)

3.2 **Construction.** The units shall be constructed in accordance with the following drawing and data lists:

SC-DL-451950 — Control, Radio Set C-4074( )/ARC-73

SC-DL-452149 — Transmitter, Radio T-879( )/ARC-73

SC-DL-452980 — Receiver, Radio R-1123( )/ARC-73

SC-DL-453400 — Mounting, MT-2699( )/ARC-73

3.3 **Preproduction Samples.** The contractor shall furnish two preproduction samples of equipment on order for approval, if required by the invitation for bids and contract. (See 4.3.)

3.4 Cleaning.

3.4.1 *Parts.* After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 **Controls.** All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 **Finish, Protective.** The units shall be given protective finish in accordance with Specification MIL-F-14072, and the applicable drawings. The final paint film shall be Color No. 37038 (Lusterless Aviation Black) according to Federal Standard No. 595. (See 4.4.)

3.7 **Marking.**

3.7.1 *General.* Marking shall conform to Specification MIL-M-13231. (See 4.4.) Front panel marking shall be Group I as described in that specification.

3.7.2 *Visibility.* Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.7.3 *Serial Numbers.* The following units shall be serial numbered:

Transmitter T-879( )/ARC-73
Receiver R-1123( )/ARC-73
Control C-4074( )/ARC-73
Mounting MT-2699( )/ARC-73

3.8 Securing of Parts.

3.8.1 *General.* Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.8.2 *Securing of parts by threaded fasteners.* Nylok or other screws with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250° F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.9 Soldering.

3.9.1 *Solder.* Solder shall conform to the equipment drawings.

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3.9.2 *Acid or Acid Salts.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for thinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and Cabling. Wiring and cabling shall be neat and sturdy.

3.10.1 *Cabling.* Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.10.2 *Slack.* Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 *Support.* Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (1) during and after subjection of the equipment to specified service conditions or (2) after service or repair of the equipment in a normal manner.

3.10.5 *Clearance.* Clearance between solder connections or bore conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 *Splicing.* Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 *Connections.*

3.10.7.1 *General.* Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bored ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other

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coating shall be applied to completed electrical connections.

3.11 **Tubes, electron; and Semiconductors (transistor and diodes).** These electron devices (tubes and semiconductors) shall conform to Specification MIL-E-1 and MIL-S-19500. (See 4.4.)

3.12 **Service Conditions** (See 4.10). The units on order shall meet the service conditions as specified herein.

3.12.1 *Receiver R-1123( )/ARC-73.*

3.12.1.1 *Temperature* (See 4.10.1). The receiver shall meet the requirements of this specification at any temperature from — 40° to + 55° C.

3.12.1.2 *Altitude* (See 4.10.1). The receiver shall meet the following limits of altitudes up to 30,000 feet above sea level:

(a) Not less than 6 db signal to noise level at 10 microvolts.

(b) Not less than 60 percent of standard test conditions audio at 20 microvolts.

(c) Not more than 6 seconds channeling time.

(d) Not more than + 10 db rise in the audio; 10 to 10,000 microvolts.

3.12.1.3 *Humidity* (See 4.10.2). After subjection to the relative humidity test the following limits shall be met:

(a) Within 15 minutes from the time primary power is applied, the sensitivity shall be not less than 25 percent of that obtained under standard test conditions.

(b) Within 4 hours from the time primary power is applied, the sensitivity shall be 75 percent of that obtained under standard test procedures.

3.12.1.4 *Vibration* (See 4.10.3). When subjected to the vibration frequencies of 10 to 55 cps the receiver shall have not less than 25 db signal plus noise to noise for 100 to 20,000 microvolts.

3.12.1.5 *Shock* (See 4.10.4).

(a) Following the application of 15G shock. the gain and sensitivity of the receiver shall be within 75 percent of that obtained prior to subjection to the shock test.

(b) Following the application of 30G shock, the receiver shall have remained in its mounting, and no part of the unit or the mounting shall have become detached and free of the table or its mounting.

3.12.2 *Transmitter T-879( )/ARC-73.*

3.12.2.1 *Temperature* (See 4.10.1). Over the range of — 40° C to + 55° C, the frequency of the rf carrier shall be within 0.01 percent of the assigned frequency. The following limits shall be met at the low and high temperatures as specified:

(a) When subjected to the low temperature the following limits shall be met:

(1) The carrier power shall be at least 70 percent of that obtainable under standard test conditions.

(2) The audio frequency distortion shall not exceed that specified in 3.14.4.

(3) The carrier noise level shall not exceed that specified in 3.14.6.

(4) All mechanical devices shall operate satisfactorily. The time required to effect a change in frequency shall not exceed that specified in 3.14.2.

(b) When subjected to high temperatures the specified limits shall be met:

(1) The audio frequency distortion shall not exceed that specified in 3.14.4.

(2) The carrier noise level shall not exceed that specified in 3.14.6.

(3) The power output shall be at least 70 percent of that obtained under standard test conditions.

(4) All mechanical devices shall operate satisfactorily.

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(5) There shall be no evidence of materials, such as grease, or potting or sealing compounds, exceeding or dripping from the unit or components.

3.12.2.2 *Altitude* (See 4.10.1). When subjected to the altitude test:

(a) The power output shall be at least 90 percent of that obtained under standard test conditions.

(b) There shall be no corona or arcing without modulation, and with the maximum level of modulation for which the unit is designed.

3.12.2.3 *Humidity* (See 4.10.2). After subjection to humidity:

(a) Immediately following the 15 minute warm-up period, the power output shall be at least 50 percent of that obtained under standard test conditions, and the output signal shall be capable of being modulated at least 50 percent by an audio signal of 1,000 cps.

(b) Within 4 hours from the time primary power is applied, the power output shall be within 90 percent to 110 percent of that obtained under standard test conditions, and the modulation capability shall be as specified in 3.14.3.

3.12.2.4 *Vibration* (See 4.10.3). When subjected to the vibration test the frequency deviation of the modulated carrier shall not exceed 1,000 cps, and the audio frequency distortion shall be as specified in 3.14.4.

3.12.2.5 *Shock* (See 4.10.4). After subjection to 15G shock:

(a) The power output shall be 90 to 110 percent of that obtained under standard test procedures.

(b) The modulation capability shall be as specified in 3.14.3.

(c) All mechanical devices shall operate satisfactorily.

(d) Following the application of 15G shocks the equipment shall remain in its mounting, and no part of the equipment or its mounting shall have become detached and free of the table or of the equipment.

3.13 **Electrical Requirements, Receiver R-1123 ( )/ARC-73** (See 4.8).

3.13.1 *Power Output, Audio* (See 4.8.1). The audio output shall be not less than 100 mw across a 500 ohm resistive load.

3.13.2 *Response, Audio* (See 4.8.2). The audio output shall not exceed +1 or —5 db from the reference value taken at 1,000 cps. The response shall decrease continuously for those frequencies below 300 cps and above 3750 cps. Above 5750 cps the level shall be not less than 20 db below the reference low above.

3.13.3 *Distortion, Audio Output* (See 4.8.3). The distortion of the audio output of the receiver shall not exceed 7 percent with an RF input of 1000 uv, 30 percent modulated at 1000 cps. The distortion of the audio output shall not exceed 20 percent with an input of 20, 1000, and 20,000 microvolts, 90 percent modulated in turn at 400, 1000, 2500 cps.

3.13.4 *Susceptibility, RF* (See 4.8.4). When an RF signal is applied between ground and the unit case, and ground and each connector pin the RF voltage required to obtain the AGC reference voltage shall be at least 46 db greater than 5 microvolts for all frequencies between 0.19 and 400 mc.

3.13.5 *Susceptibility, Audio* (See 4.8.5). When an AC voltage is introduced into the primary power leads, the attenuation shall be not less than 30 db between 30 and 20,000 cps.

3.13.6 *VSWR, Receiver* (See 4.8.6). The receiver voltage standing wave ratio on the transmission line shall not exceed a value of 4:1 over the radio frequency range from 108-151.95 mc.

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3.13.7 *Spurious Response* (See 4.8.7). The spurious response input for the referenced output shall be at least 80 db including images.

3.13.8 *Emission of Radio-Frequency* (See 4.8.8). The emission of radio frequency energy from the receiver over the frequency range of .190 to 1500 mc shall not exceed 200 microvolts between any two cable terminals or any terminal and ground, and 400 microvolts across the test antenna, or, in any case where the receiver is designed for a specific source impedance, across a resistive load equal to the source impedance for which the equipment is designed.

3.13.9 *Selectivity, Overall* (See 4.8.9). The frequency shift with 6 db attenuation shall be not less than ± 20 kc, and with 60 db attenuation shall be not more than 40 kc. Asymmetry at 3 db, 20 db, 40 db, 60 db, and 80 db shall be not more than 15 percent.

3.13.10 *Sensitivity* (See 4.8.10). The signal plus noise to noise ratio shall be not less than 6 db for a 3 microvolt input.

3.13.11 *Cross Modulation* (See 4.8.11). The cross modulation output with 1000 microvolts and 20.000 microvolts (alternately) RF input, and modulated 30 percent, at 400 cps, ± 50 kc off-resonance, shall be at least 10 db below the audio output obtained with a 20 microvolt RF input, and modulated 30 percent at 400 cps at resonance. With 20,000 uv RF input, and modulated 30 percent, at 400 cps, ± 100 kc off-resonance, the cross modulation output shall be at least 10 db below the audio output obtained with a 2000 microvolt RF input modulated 30 percent, 400 cps resonance.

3.13.12 *Channel Selection* (See 4.8.12). The longest cycling time of the frequency selector system shall not exceed 6 seconds.

3.13.13 *AGC Operation* (See 4.8.13). The receiver output shall not vary more than 3 db with an input signal from 5 microvolts to 100,000 microvolts. Variations of percentage modulation shall have negligible effect on the automatic gain control.

3.13.14 *ATC Output* (See 4.8.14). The output circuit having a nominal impedance of 1000 ohms (exclusive of the effect of coupling condenser) when operated into an external load impedance of 250,000 ohms shall have a frequency response from 300 cps to 25 kc within 6 db of the reference level at 1000 cps, and shall supply an open circuit voltage output of 0.5 rms with an input signal of 1000 microvolts modulated at 30 percent at 1000 cps. This output shall not pass through the noise limiter.

3.13.15 *Receiver Disable Control* (See 4.8.15). With the application of + 27.5 volts dc to pin 23, the receiver shall be disabled.

3.13.16 *Squelch Control* (See 4.8.16). The carrier operated noise silencer shall mute the receiver output until application of any preset input from 3 microvolts to 25 microvolts.

3.14 Electrical Requirements, Transmitter T-879( )/ARC-73 (See 4.9). The transmitter shall conform to the requirements specified herein.

3.14.1 *Power Output, RF* (See 4.9.1). When operated at rated input power for 15 minutes, the carrier power output measured into a 52 ohm resistive load shall be a minimum of 25 watts in the range of 116.00-133.95, and 20 watts in the range of 133.95 through 149.95 megacycles.

3.14.2 *Channel Selection* (See 4.9.2). The longest time required for the transmitter to effect a change from the longest channeling cycle shall not exceed 6 seconds.

3.14.3 *Modulation Capability* (See 4.9.3). The transmitter carrier shall be capable of being amplitude modulated not less than 85 percent at any of the channel frequencies.

3.14.4 *Audio Frequency Distortion* (See 4.9.4). The combined distortion and noise in

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the demodulated output of the transmitter shall not exceed 25 percent of the total demodulated output of the modulated frequencies of 400, 1000 and 2500 cps, when the level of the audio input to the transmitter is held constant at the value producing at least 85 percent modulation at 1000 cps.

3.14.5 *Modulation Fidelity* (See 4.9.5). The percentage of modulation of the output carrier shall not vary more than 6 db when the frequency of the audio input signal is varied over the range of 350 to 2500 cps and the level of the audio input signal held constant at a prescribed value. The prescribed value shall be noted at the frequency of maximum response which produced 85 percent modulation or the value immediately below that at which clipping or limiting action is present.

3.14.6 *Carrier Noise Level* (See 4.9.6). The demodulated noise in the transmitter output shall be at least 35 db below the demodulated output obtained when the output signal is modulated 85 percent at 1000 cps.

3.14.7 *Sidetone* (See 4.9.7). With sidetone output, the audio-frequency response shall not vary more than 10 db over the range of 350 to 2500 cps when the level of the audio input to the transmitter is held constant at a prescribed value. The prescribed value shall be that noted at the frequency of maximum response which produces 85 percent modulation or the value immediately below that at which clipping or limiting action is present.

3.14.8 *Field Servicing Controls* (See 4.9.8). Operating the meter switch through all its positions shall indicate on the meter that the voltage or current is present in all circuits for proper tuning and operation of the transmitter.

3.14.9 *Susceptibility, RF* (See 4.9.9). The residual noise on the carrier shall be at least 30 db below the level of the modulated carrier.

3.14.10 *Spurious Radiation* (See 4.9.10). Radiation from the transmitter for voice modulation at frequencies outside the band ± 25 kc from the assigned carrier frequency, when the peak level of a voice signal input to the transmitter is at least 10 db above that which produces 85 percent modulation, shall be as follows:

(a) The transmitter output power on discrete frequencies outside the bands from ± 25 kc from the assigned carrier frequency, and from — 25 kc to — 50 kc from the assigned carrier frequency shall not exceed 125 milliwatts.

(b) The transmitter output power on discrete frequencies outside the band from ± 50 kc from the assigned carrier frequency shall not exceed 50 milliwatts.

3.15 Remote Control C-4074( )/ARC-73 (See 4.14). The remote control unit shall select the operating frequencies of the transmitter and receiver as well as control the system On-Off, squelch, and volume control.

3.16 Mounting MT-2699( )/ARC-73 (See 4.10.4). This unit shall provide for mounting of Transmitter T-879( )/ARC-73 and Receiver R-1123( )/ARC-73. The mounting shall meet the service condition requirements with the receiver and transmitter.

3.17 Voltage Variations (See 4.12). The injection frequency of the equipment shall not deviate more than ± 0.0035 percent from the frequencies specified, when the input voltage is changed from 24 to 29 volts dc.

3.18 Low Voltage (See 4.11). When the primary power voltage is 80 percent of standard test voltage the units shall operate electrically and mechanically and meet the requirements of this specification. Gradual reduction of the primary power voltage from 80 percent to 50 percent of standard test voltage shall produce no condition detrimental to the reliability of the units. Reduction of the primary power voltage from 50 percent to zero percent of the standard test voltage shall produce no evidence, external to the units, of the presence of fire or smoke.

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3.19 **System Operation** (See 4.14). The units shall be interconnected and shall be operated at a duty cycle of ¼ minute transmit and ¾ minute receiving.

3.20 **Transient Protection** (See 4.13). The transient protector circuit in the receiver and transmitter power supplies shall operate between 32 and 35 volts.

3.21 **Preconditioning** (See 4.5 and 4.6).

3.21.1 *Vibration Preconditioning* (See 4.6.1). The units shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the vibration preconditioning test for two minutes.

3.21.2 *Electrical Preconditioning* (See 4.6.2). The unit shall be capable of meeting the inspection of Section 4 after subjection to the electrical preconditioning test.

3.22 **Operational test (500 hours)** (See 4.16). The units shall be capable of 500 hours of continuous operation at a 10:1 receive-transmit ratio and 29 V dc input voltage, with no replacement of parts or realignment other than adjustment of control unit controls. During and at the conclusion of the operation test, the units shall meet the specified tests. The duty cycle shall be 23 hours on and one hour off.

3.23 **Interchangeability** (See 4.18). Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the units. Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 **Operation Inspection** (See 4.15). Remote Control C-4074( )/ARC-73 shall be matched with Receiver R-1123( )/ARC-73 and Transmitter T-879( )/ARC-73. Operation of frequency selectors, switches and controls shall be verified.

3.25 **Technical Literature and Spare Parts.** Technical literature and spare parts shall be furnished as specified in the contract. Spare parts shall be identical to corresponding spare parts in the equipment furnished on the order.

3.26 **Workmanship.** The units shall be manufactured and assembled in accordance with the applicable drawings and the following paragraphs:

Cleaning	3.4
Controls	3.5
Securing of Parts	3.8
Soldering	3.9
Wiring and Cabling	3.10

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for Inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Classification of Inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Inspection covered by subsidiary documents (See 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.19).

4.3 **Preproduction Inspection.** This inspection will be performed by the Government unless otherwise specified in the contract. If

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shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C, (see Tables II, III, and IV, respectively). The preproduction inspection will normally be performed in this order: (1) vibration, and (2) shock; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE I. *Preproduction Inspection*

Inspection	Req. Par. Receiver	Transmitter	Insp. Par.
Altitude	3.12.1.2	3.12.2.2	4.10.1
Humidity	3.12.1.3	3.12.2.3	4.10.2
Shock	3.12.1.5	3.12.2.5	4.10.4
Low Voltage	3.18	3.18	4.11

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of the units before preparation for delivery:

Item	Where Required
Finish	3.6
Marking	3.7
Electron tubes and semiconductors	3.11

4.5 Quality Conformance Inspection of Equipment before Preparation for Delivery. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government — at its discretion — may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. All units on order shall be subjected to preconditioning (See 3.21) after final assembly. The preconditioning (3.21) shall be performed once, except when the units require reprocessing.

4.5.1 *Group A Inspection.* This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.15) shall be last. (See 6.5).

TABLE II. *Group A inspection*

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.26	4.17		
Receiver R-1123( )/ARC-73				
RF and Variable IF A1			0.65%	2.5%
500 kc IF and Audio A2			0.65%	2.5%
DC Power Supply A3			0.65%	2.5%
500 kc Passive Filter A5			0.65%	2.5%
Chassis Module A6			0.65%	2.5%
R-1123( )/ARC-73 (Assembly)			1.0%	4.0%
Visual and Mechanical	3.26	4.17		
Transmitter T-879( )/ARC-73:				
Transmitter chassis and front panel			0.65%	2.5%
Front panel terminal board			0.65%	2.5%
RF module A1			0.65%	2.5%
Modulator A2			0.65%	2.5%
DC power supply A4			0.65%	2.5%
Control C-4074( )/ARC-73			2.5 dphu	10.0 dphu
Mount MT-2699( )/ARC-73			1.0%	4.0%

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TABLE II. *Group A inspection* (Continued)

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Electrical				
Receiver R-1123( )/ARC-73:				
Power output, audio	3.13.1	4.8.1	1.0% for* the combined group	
Response, audio	3.13.2	4.8.2		
Distortion, audio output	3.13.3	4.8.3		
Selectivity, overall	3.13.9	4.8.9		
Sensitivity	3.13.10	4.8.10		
Cross modulation	3.13.11	4.8.11		
Channel selection	3.13.12	4.8.12		
AGC operation	3.13.13	4.8.13		
Receiver disable control	3.13.15	4.8.15		
Squelch control	3.13.16	4.8.16		
Transmitter T-879( )/ARC-73:				
Power output, RF	3.14.1	4.9.1	1.0% for* the combined group	
Channel selection	3.14.2	4.9.2		
Modulation capability	3.14.3	4.9.3		
Audio frequency distortion	3.14.4	4.9.4		
Modulation fidelity	3.14.5	4.9.5		
Carrier noise level	3.14.6	4.9.6		
Sidetone	3.14.7	4.9.7		
Field service control	3.14.8	4.9.8		
Transient Protection	3.20	4.13	1.0%	
Operational	3.15, 3.24	4.15	0.1%	

DPHU = Defects per hundred units.
* All electrical defects are considered major.

4.5.2 *Group B Inspection.* This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and have met the Group A inspection.

4.5.3 *Order of Inspection within Group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. *Group B Inspection*

Inspection	Rqt. Para.	Insp. Para.	AQL
Receiver R-1123(|)/ ARC-73:			
VSWR receiver	3.13.6	4.8.6	6.5%
ATC output	3.13.14	4.8.14	6.5%
Voltage variations	3.17	4.12	6.5%
Transmitter T-897( )/ ARC-73:			
Voltage variation	3.17	4.12	6.5%
AN/ARC-73			
Interchangeability (dimensions)	3.23	4.18	6.5%
Interchangeability (module performance)	3.23	4.18	6.5%

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4.5.4 *Group C Inspection.* This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

TABLE IV. *Group C Inspection*

Inspection	Rqt. Para.	Insp. Para.
Subgroup I		
AN/ARC-73( )		
Temperature	3.12.1.1, 3.12.2.1	4.10.1
Vibration	3.12.1.4, 3.12.2.4	4.10.3
System operation	3.19	4.14
Subgroup II		
Receiver R-1123( )/ ARC-73		
Susceptibility, RF	3.13.4	4.8.4
Susceptibility, audio	3.13.5	4.8.5
Spurious response	3.13.7	4.8.7
Emission of radio frequency	3.13.8	4.8.8
Transmitter T-879( )/ ARC-73		
Susceptibility, RF	3.14.9	4.9.9
Spurious radiation	3.14.10	4.9.10
Operational Test (500 hr)	3.22	4.16

4.5.4.1 *Sampling for Inspection of Equipment.* Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.4.1.2 *Subgroup I.* For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections. two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.4.1.3 *Subgroup II.* For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 350, or two each three months if less than 350 are produced, shall be selected.

4.5.4.1.4 *Noncompliance.* If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of production which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4).

4.5.5 *Reinspection of Conforming Group B and Group C Sample Units.* Unless otherwise specified, sample units which have been subjected to and have passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 **Preconditioning Tests.**

4.6.1 *Vibration Preconditioning Test.* Place the units, with isolators (if any) removed or blocked, in its normal operating position on a vibration table. The vibration system shall operate at 55 cps vibration giving a vertical excursion sufficient to maintain an acceleration of 5 G's.

4.6.2 *Electrical Preconditioning (20 hour run-in).* Each Transmitter T-879( )/ARC-73 and each Receiver R-1123( )/ARC-73 shall be electrically preconditioned by subjecting them to a 20 hour run-in period. For this preconditioning a test fixture, the design and performance of which shall be approved by the contracting officer, may be used. The T-879( )/ARC-73 and R-1123( )/ARC-73, connected to the approved test fixture. shall be operated continuously for 20 hours without failure. The T-879( )/ARC-73 shall transmit for 15 seconds, and the R-1123( )/ARC-73 shall receive for 45 seconds. Frequency selection shall be performed once every 15 minutes for at least 6 channels during the run-in period. The transmitter shall be terminated in a 52 ohm dummy load.

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4.7 **Test equipment.** The principal units of test equipment required for the electrical tests are listed below. Unless otherwise specified in the bid request and contract, the test equipment shall be furnished by the contractor.

Power Supply	15 amp	28 V dc Behlman Invar TPA-15 or equal
RF Wattmeter	0.20-80 watts	50 ohm Bird #612 or equal
Grid Dip Meter	10-450 mc	Measurement Corp #59VHF or equal
Audio Power Meter	20 cps-20 kc 0.1-5 watts	General Radio #1840 or equal
Distortion analyzer	20 cps-20 kc	Hewlett-Packard #3308 or equal
AC VTVM	.001V-300V	Hewlett-Packard #400D or equal
Audio oscillator	20 cps-600 kc	Hewlett-Packard #200CD or equal
VTVM	IV-1000 VDC	Hewlett-Packard #410B or equal
VHF detector	10-500 MC	Hewlett-Packard #417A or equal
RF generator	10 Mc-420 Mc	Hewlett-Packard #608D or equal
Electronic counter	10 cps 500 mc	Hewlett-Packard #5245L, 5253A or equal
RF bridge	2-2000 ohms 52-500 mc	Hewlett-Packard #803A or equal
Peak AC VTVM		Ballantine #305A or equal
Oscilloscope preamp	DC-15 Mc	Tektronix #531A, CA or equal
Test Set, Radio AN/ ARM-45( )		GLE or equal

4.7.1 *Test Conditions.* Unless otherwise stated herein the following test conditions shall apply:

(a) Temperature—Prevailing room ambient.

(b) Pressure — Prevailing room ambient.

(c) Humidity — Prevailing room ambient.

(d) Input voltage — 27.5 ± 0.5 V dc.

4.8 **Receiver R-1123( )ARC-73, Electrical.**

4.8.1 *Power Output, Audio.* A 3 microvolt signal modulated 30 percent at a 1000 cps shall be fed into the antenna input connector of the receiver, over the frequency range of 108 to 151.95 mcs. The equipment shall meet the requirement of 3.13.1.

4.8.2 *Response, Audio.* A 1,000 microvolt 127.5 mc signal shall be fed to the antenna connector of the receiver. The audio modulation shall be held constant at 30 percent and the frequency varied from 200 to 5750 cps. The audio output of the receiver shall be adjusted to 50 mw. The receiver shall meet the requirements of 3.13.2.

4.8.3 *Distortion, Audio Output.* A modulated signal shall be fed to the antenna connector of the receiver at a frequency of 127.5 mc. The audio output into a 500 ohm load shall be adjusted and held to 100 mw. The total harmonic distortion shall be measured with a noise and distortion analyzer. The receiver shall meet the requirements of 3.13.3.

4.8.4 *Susceptibility, RF.* The receiver positioned to 127.5 mc and the AGC measured for a 5 microvolt signal shall be fed into the antenna connector. Squelch control shall be set for a maximum sensitivity. The antenna signal shall be removed and a 1000 microvolt signal shall be applied between ground and the equipment case, and in turn between ground and each connector pin. A line isolator, as specified in Specification MIL-I-6181 shall be inserted in each lead. The AGC output for all interfering signals between .19 mc and 400 mc shall meet the requirements of 3.13.4.

4.8.5 *Susceptibility, Audio.* Apply a 100 microvolt signal to the receiver input and adjust the audio gain for 100 milliwatts with a 30 percent modulation. With the signal unmodulated, apply two volts of AC to the DC power input. The receiver shall meet the requirements of 3.13.5.

4.8.6 *VSWR, Receiver.* The input impedance shall be measured at frequencies from 108 to 151.95 mc, and shall meet the requirements of 3.13.6.

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4.8.7 *Spurious Response.* An RF signal, modulated 30 per cent at 1000 cps, shall be fed into the antenna connector of the receiver. The input level shall be adjusted to 2 microvolts at various frequencies, and a reference AVC and audio reading shall be noted. The input level shall be raised to 100 db above 2 microvolts and the frequency varied from 0.190 to 1500 mc. Record frequencies where responses are noted. The receiver shall meet the requirements of 3.13.7.

4.8.8 *Emission of Radio Frequency.* A line isolator shall be inserted in each power and control lead, as indicated in Specification MIL-I-6181. All signals present at the antenna connector of the receiver shall be measured with a noise and field strength meter at frequencies from 108 to 151.95 mc. Equipment shall meet the requirements of 3.13.8.

4.8.9 *Selectivity, Overall.* The receiver shall be channeled to 127 mc and the 127 mc signal at 3 microvolts shall be fed into the antenna. (Note AVC level.) The signal level shall be increased 3, 6, 20, 40, 60, and 80 db, and at each level the frequency shall be shifted off resonance an amount to bring the AVC back to its reference level. (Note: Allowance shall be made for crystal tolerance.) The frequency shift on either side shall meet the requirements of 3.13.9.

4.8.10 *Sensitivity.* The signal plus noise to noise ratio shall be taken as the ratio of audio output with a 1000 cps, 30 percent modulated input, to the audio output with an unmodulated input. The ratio shall be measured at the frequencies from 108 to 151.95 and shall be as specified in 3.13.10.

4.8.11 *Cross Modulation.* Two signal generators shall be combined through a "T" pad to the antenna connector of the receiver. Signal generator #1 shall be tuned to the test frequency and the receiver output adjusted to a convenient level with 20 microvolts (modulated) input. Record the level. Signal generator #1 shall remain in operation but modulation shall be removed. Then signal generator #2 shall be placed in operation ±50 kc off-resonance with an input level of 1000 microvolts and 20,000 microvolts alternately (modulated). Record measurements. The receiver shall meet the requirements of 3.13.11.

4.8.12 *Channel Selection.* After a warm-up period of one minute the frequency selection time shall be measured over the longest tuning cycle. The longest channeling time shall not exceed the requirements of 3.13.12.

4.8.13 *AGC Operation.* After a warm up period of one minute perform the following:

(a) Set the receiver and signal generator Hewlett Packard (608D or equal) to 127.5 mc.

(b) Set the signal generator for an output of 5 microvolts modulated 30 percent at 1000 cps. Adjust the generator frequency for maximum AVC voltage.

(c) Adjust the Aural Level control for a 20 milliwatt (0 reference) reading on the output power meter.

(d) Adjust the signal generator for an output of 100,000 microvolts and record the db change in the audio output from a zero reference level. The requirement of 3.13.13 shall be met.

4.8.14 *ATC Output.* With an input voltage of 27.5 V dc, feed a 1000 microvolt 117.50 mc signal to the antenna connector of the receiver. Hold the audio modulation constant at 30 percent and vary the frequency from 300 to 25,000 cps. The ATC output of the receiver shall be set at maximum. The requirements of 3.13.14 shall be met.

4.8.15 *Receiver Disable Control.* Feed an RF signal at 127.50 mc, modulated 30 percent at 1000 cps to the antenna connector of the receiver. Audio output shall be adjusted to 100 mw across a 500 ohm load. With the application of 27.5 V dc to pin 23 of the receiver rear plug (P1), the requirements of 3.13.15 shall be met.

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4.8.16 *Squelch Control.* Set the receiver and signal generator to 127.5 mc and turn the remote control unit squelch control fully counterclockwise. Set the generator output level to 3 microvolts modulated 30 percent by 1000 cps and adjust the receiver squelch control so that squelch just opens. Increase the generator output level at this point. Slowly increase the generator output level until the squelch just opens. Record generator level at this point. Increase the signal generator modulation to 90 percent and record the output level need to just open the squelch. Decrease the signal generator modulation to 10 percent and record the output level needed to just open the squelch. Turn the receiver squelch control fully counter-clockwise and note that the squelch closes. Push the Receiver Squelch Disable button and the squelch shall open. Set the signal generator for 30 percent modulation at 1000 cps and increase the generator output level until the squelch opens. Record the output level at this point. Turn the remote control squelch control fully clockwise and record the generator output level required to just open the squelch. The requirements of 3.13.16 shall be met.

4.9 Transmitter T-879( )/ARC-73.

4.9.1 *Power Output, RF.* Using Test Set, Radio, AN/ARM-45, or equal, Channel the transmitter to 116 mc and set the transmitter standby switch on the test set to transmit position. With the primary dc voltage adjusted to 27.5V dc at the input of the transmitter record the unmodulated output power. Channel the transmitter from 116 to 149 mc and measure the power output at each megacycle position. Select one of the megacycle channels that indicates the lowest power reading and channel the transmitter to each of the 50 kc positions, and measure the output power. Requirements of 3.14.1 shall be met.

4.9.2 *Channel Selection.* Using a stop watch, or oscilloscope, the time required for the transmitter to be channeled from the moment the frequency selector unit is manipulated until the transmitter is on the new frequency shall be measured. The requirements of 3.14.2 shall be met.

4.9.3 *Modulation Capability.* With a sine wave applied to the audio input of 0.1 volts and the dc supply adjusted to 27.5 ± 0.5 volts, the modulation gain control shall be capable of adjusting to determine compliance with 3.14.3.

4.9.4 *Audio Frequency Distortion.* With a sine wave (1 kc) applied to the audio input of 0.75 V (RMS) and the modulation level adjusted to 85 percent modulation, the requirements of 3.14.4 shall be met.

4.9.5 *Modulation Fidelity.* With the equipment in transmit condition vary the audio oscillator frequency between 350 and 2500 cps at a constant input voltage level and observe the frequency which produces a maximum modulation percentage of the carrier. Adjust the audio output at this frequency to produce 85 per cent modulation or to a level just below that at which clipping or limiting action takes place. Note the demodulated voltage level for zero reference value. Adjust the audio oscillator for 350 cps at the audio input reference obtained above. Measure the db change from the demodulated zero reference level. Repeat the procedure using audio frequencies of 2500 cps and 1000 cps and measure the db change from the demodulated output level. Adjust the audio oscillator output for 0.75 volts at 1000 cps. Note the demodulated voltage reference level. Requirements of 3.14.5 shall be met.

4.9.6 *Carrier Noise Level.* During the above test, measure the transmitted noise level as indicated on an AC voltmeter. Record the db difference between the 1000 cps demodulated reference level and the noise level. The requirmeents of 3.14.6 shall be met.

4.9.7 *Sidetone.* With the equipment in transmit condition adjust the audio oscillator output for 0.75 volts at 1000 cps. Adjust the sidetone level control (R-1) on the transmitter main frame, clockwise to obtain 7.1V ac reference sidetone output on an AC voltme-

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ter. Lock the R-1 at this level. Set the audio oscillator output for 7.5 volts at 350 cps and measure the db change in the sidetone output level. Vary the frequency of the audio oscillator between 350 and 2500 cps while maintaining the input voltage level constant at 0.75 volts and record the maximum sidetone output variation from the zero db reference level. The requirements of 3.14.7 shall be met.

4.9.8 *Field Service Control.* After a one minute warm-up period, adjust the dc supply to 27.5V ±.05V. Adjust the audio oscillator to 1 kc. Adjust output for 85 percent modulation of transmitter carrier. Switch the meter through all positions. The requirements of 3.14.8 shall be met.

4.9.9 *Susceptibility, RF.* A 1,000 microvolt signal, modulated 30 percent at 1000 cps, and varied over the frequency range of 118 to 152 mcs, shall be introduced into the rear plug primary power and ATCSS lines. The effect shall be monitored visually and aurally on the side tone output and noise level of the carrier. Requirement of 3.14.9 shall be met.

4.9.10 *Spurious Radiation.* Radiations shall not exceed the requirements of 3.14.10.

4.10 Service Conditions.

4.10.1 *Temperature and Altitude Test.* The units mounted on MT-2699( )/ARC-73 shall be subjected to the temperature and altitude test of Specification MIL-T-5422 for Class 1A equipment. The requirements of 3.12.1.1, 3.12.2.1, 3.12.1.2 and 3.12.2.2 shall be met.

4.10.2 *Humidity.* The complete equipment shall be placed non-operative in a climate chamber with relative humidity of between 95 and 100 percent, and a temperature of 50° C. for a 48 hour period. During this period the humidity shall be raised once an hour to make the chamber "rain." After this period the equipment shall be removed from the chamber and condensed moisture wiped off. The primary voltage shall be applied within 5 minutes of removal from the chamber and the equipment given a 15 minute warm up period. The equipment shall meet the requirements of 3.12.1.3 and 3.12.2.3.

4.10.3 *Vibration.* The receiver and transmitter on MT-2699( )/ARC-73 and the remote control shall be subjected to the vibration test of Specification MIL-T-5422 for compliance with Curve IV, figure 2, in that specification.

4.10.4 *Shock.* The receiver and transmitter on MT-2699( )/ARC-73 and the remote control shall be attached to shock table in a normal operating position. Sensitivity and power output shall be checked for a 30 percent 1000 cps modulated signal applied to the antenna connector. With the equipment not operating, three shocks in each of the following planes at an acceleration of 15G and a duration of 10 milliseconds shall be applied:

(a) Normal upright.

(b) Suspended upside down.

(c) Major horizontal axis 45 degrees with the plane of the table.

(d) Minor horizontal axis 45 degrees with the plane of the table.

The requirements of 3.12.1.5 and 3.12.2.5 shall be met.

4.11 Low voltage. Turn on equipment and allow to warm up for one minute. Adjust the supply voltage to 22 V dc (80 percent of normal). The units shall be operated with primary voltages varied as indicated and shall meet the requirements of 3.18.

4.12 Voltage variation. Set the megacycle and kilocycle controls to 108.00 mc. Record each 0.05 megacycle step through 0.95 megacycles. Set the control for 108.00 mc. Record each even megacycle through 151.00 mc. The units shall meet the requirement of 3.17.

4.13 Transient protection. Connect an oscilloscope between collector terminal of Q2 and Q21 on the front panel of the transmit-

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ter and an oscilloscope to the junction of A3R7 and A3C2 and ground. The transient generator, or equal, shall be discharged at 32, 33, 34, and 35 volts for compliance with 3.20.

4.14 **System Operational Test.** The Radio Set AN/ARC-73 shall be tested on at least 12 channels between 108.00 and 150.00 mc, transmitting and receiving one each channel. Frequency measurements shall be made on at least 4 operating channels, and power output measured on at least 15 operating channels. Test shall be made using 24.0 V dc.

4.15 **Operational Test.** Each unit of Radio Set AN/ARC-73 shall be combined with its corresponding components to form an operational unit. (Test Set, Radio, AN/ARM-45 or equal may be substituted with the approval of the contracting officer.) Satisfactory transmission and reception quality shall be checked on at least 9 channels and shall indicate proper operation of the equipment.

4.16 **Operational Test (500 hour).** The units shall be subjected to the following operating conditions at an ambient temperature of 70° F:

(a) During the 500 hour operation the following test shall be performed:

(1) Channel selection shall be performed 230 times during each 23 hours of operation; 10 seconds shall be allowed between each selection.

(2) A daily check on power output and sensitivity at each end and middle of each band of the units. Degradation shall not exceed 30 percent from the limits specified in 3.14.1 and 3.13.10.

(b) At the conclusion of the 500 hour period perform the following tests:

Receiver		
Audio output	3.13.1	4.8.1
Audio distortion	3.13.3	4.8.3
Selectivity	3.13.9	4.8.9
Squelch	3.13.16	4.8.16
Sensitivity	3.13.10	4.8.10
Transmitter		
RF power output	3.14.1	4.9.1
Modulation capability	3.14.3	4.9.3
Audio frequency distortion	3.14.4	4.9.4
Carrier noise level	3.14.6	4.9.6

4.17 **Visual and Mechanical Inspection.** The equipment shall be inspected for the defects as listed in Standard MIL-STD-252.

4.18 **Inspection for Interchangeability.** The mechanical dimensions and electrical performance required by the mechanical and electrical test fixture drawings shall be measured to determine conformance with the physical and functional interchangeability requirements of 3.23. When a listed dimension or performance requirement is not within specified limits it shall be considered a major defect. The following dimensions or performance shall be gaged or measured:

4.18.1 *Mechanical (Dimensions).* The gage points covered by the drawings listed on the following gage list shall apply: SC-GL-323057, Mechanical Gages for Radio Set AN ARC-73( ).

4.18.2 *Electrical Performance (Modules).* Prior to assembly into the units, modules shall be tested to determine compliance with the performance requirements specified in the following gage lists and drawings:

(a)	SC-GL-323019	Test Fixture for RF and IF Amplifier Module A1 (Receiver)
(b)	SC-GL-323020	Test Fixture for IF and Audio Module A2 (Receiver)
(c)	SC-GL-323021	Test Fixture for RF Unit Module A1 (Transmitter)
(d)	SC-GL-323022	Test Fixture for Modulation Module A2 (Transmitter)
(e)	SM-D-462262	Power Supply Module A-4 (Transmitter)

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(f) SM-D-453349 Power Supply Module A-3 (Receiver)

4.19 **Quality Conformance Inspection of Preparation for Delivery.** Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.20 **Rough Handling Test (Preparation for Delivery).** When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2(e)), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling: 4.15 Operational Test.

5. PREPARATION FOR DELIVERY

5.1 Preparation and Packaging.

5.1.1 *Level A.* Radio Set AN/ARC-73( ) and individual units shall be packaged as follows:

5.1.1.1 *Technical Manual.* Each technical manual shall be packaged Method 1C-6 as follows: Inclose each technical manual within a close-fitting type II, class d bag conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.1 *Radio Set AN/ARC-73( ). Individual Units.*

5.1.1.3 *Transmitter T-879( )/ARC-73.* Each transmitter shall be packaged Method 11B as follows: Cushion each transmitter on all surfaces with cells or pads or both, fabricated of corrugated fiberboard. Type CF, class domestic, Variety SW, Grade 175 conforming to Specification PPP-F-320, designed to absorb the shock of impact normally encountered in handling and transit. Place the cushioned transmitter together with the required number of units of activated dessicant within a close-fitting fiberboard box conforming to Specification PPP-B-636, type 1, class L. The dessicant shall conform to Specification MIL-D-3464, in unit quantities required by the formula in Specification MIL-P-116. The dessicant shall be placed within the cushioning cells. A humidity indicator card conforming to Military Standard MS-20003 shall be secured with pressure sensitive tape conforming to Specification PPP-T-76 to the underside of one of the top inner flaps of the container immediately inclosing the contents. Box closure shall be in accordance with the appendix of the referenced box specification. The box corners shall be blunted and any stapled or stitched joints shall be covered with gummed paper tape conforming to Specification UU-T-111. The sealed interior box shall be placed within a water-vaporproof barrier bag conforming to the requirements of Specification MIL-B-117 for type 1, class e, and fabricated with one (1) dimension to allow three (3) resealings for inspection purposes. Heat seal the water-vaporproof barrier bag leaving a small opening to exhaust the air. Exhaust the excess air and complete the seal. The time lapse between positioning the dessicant and the completion of the barrier bag seal must be held to an absolute minimum. Place the water-proofed Transmitter within a close-fitting fiberboard box, type 1, class 2, W5c, Style RSC conforming to Specification PPP-B-636. Any stapling on the inside of the exterior box shall be covered with water-resistant, pressure sensitive tape conforming to Specification PPP-T-76, to prevent rupturing the barrier bag. Closure will be in accordance with the appendix of the box specification.

5.1.1.4 *Receiver R-1123( )/ARC-73.* The Receiver shall be packaged Method 11B following the procedures specified in paragraph 5.1.1.3.

5.1.1.5 *Control Unit C-4074( )/ARC-73.* The Control shall be packaged Method 11b following the procedures specified in paragraph 5.1.1.3.

5.1.1.6 *Mounting MT-2899( )/ARC-73 with Hardware and Plugs.* Each Mounting with three sets of hardware and plugs shall

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be packaged Method 1C5 as follows: Inclose hardware and plugs within a cotton bag conforming to Specification PPP-B-20. Secure bag of hardware and plugs to the mounting with pressure sensitive tape conforming to Specification PPP-T-76. Cushion each mounting on all surfaces with cells or pads or both fabricated of fiberboard, Type CF, Variety SW, Grade W5c, conforming to Specification PPP-F-320. Place the cushioned mounting within a close-fitting fiberboard box Type 1, class 2, Grade W5, style RSC, conforming to the requirements of Specification PPP-B-636. Close box by securing the top and bottom inner and outer flaps with water-resistant adhesive conforming to the requirements of Specification MIL-A-101 and applied to all areas of contact between the flaps. Alternately the bottom inner and outer flaps may be secured by stitching or stapling as prescribed in the appendix of Specification PPP-B-636, with the top inner and outer flaps secured with water-resistant adhesive as described above. In addition, seal the top and bottom center seams and edges and the manufacturer's joint with water-resistant pressure-sensitive tape conforming to the requirements of Specification PPP-T-76.

5.1.1.7 *Radio Set AN/ARC-73.* The Radio Set shall be packaged in accordance with paragraphs 5.1.1.3, 5.1.1.4, 5.1.1.5, and 5.1.1.6.

5.1.2 *Level C.* Radio Set AN/ARC-73( ) and individual units shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity (see 6.8).

5.1.3 *Package Performance.* Package testing shall be in accordance with the requirements of Specification MIL-P-116. The rough handling and functional test will be required only when invoked in the bid request or contract (see 6.2).

5.2 **Packing.**

5.2.1 *Radio Set AN/ARC-73( ).* The Radio Set and individual units shall be packed as follows:

5.2.1.1 *Radio Set AN/ARC-73( ) Individual Units.*

5.2.1.2 *Transmitter T-879( )/ARC-73.* Place a quantity of transmitters packaged as specified in 5.1.1.3 within a triple-wall corrugated fiberboard box, class 2, Grade A, Style E, conforming to Specification PPP-B-640. Closure shall be in accordance with the appendix of the applicable box specification. Gross weight of the shipping containers shall not exceed 200 pounds.

5.2.1.3 *Receiver R1123( )/ARC-73.* Place a quantity of receivers packaged as specified in 5.1.1.4, and continue with procedures specified in 5.2.1.2.

5.2.1.4 *Control Unit C-4074( )/ARC-73.* Place a quantity of control units packaged as specified in 5.1.1.5, and continue with procedures specified in 5.2.1.2.

5.2.1.5 *Mounting MT-2699( )/ARC-73.* Place a quantity of Mountings packaged as specified in 5.1.1.6 and continue with procedures specified in 5.2.1.2.

5.2.1.6 *Radio Set AN/ARC-73( ).* Place the Radio Set AN/ARC-73( ) packaged as specified in 5.1.1.7 within a class 2, style 4 nailed wood box, or class 2, Type III, Style A cleated plywood box or triple-wall corrugated fiberboard box, class 2, Grade A, Style E conforming to Specification PPP-B-621, PPP-B-601 or PPP-B-640, respectively. Fabricate the box to fit contents snugly. Place the technical manual packaged as specified in 5.1.1.1 on top of the contents directly under lid of the box. Closure shall be in accordance with the appendix of the applicable box specification.

5.2.2 *Level B.* Radio Set AN/ARC-73 and individual units shall be packed as follows:

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5.2.2.1 *Radio Set AN/ARC-73( ) Individual Units.*

5.2.2.2 *Transmitter T879( )/ARC-73.* The transmitter shall be packed as specified in 5.2.1.2 except that the triple-wall fiberboard box shall be class 1. The box closure shall be as specified in the appendix of the applicable box specification.

5.2.2.3 *Receiver R-1123( )/ARC-73.* The receiver shall be packed as specified in 5.2.1.3 except that the triple-wall fiberboard box and closure shall be as specified in 5.2.2.2.

5.2.2.4 *Control Unit C-4074( )/ARC-73.* The control shall be packed as specified in 5.2.1.4 except that the triple-wall fiberboard box and closure shall be as specified in 5.2.2.2.

5.2.2.5 *Mounting MT-2699( )/ARC-73.* The mounting shall be packed as specified in 5.2.1.5 except that the triple-wall fiberboard box and closure shall be as specified in 5.2.2.2.

5.2.2.6 *Radio Set AN/ARC-73( ).* The radio set shall be packed as specified in 5.2.1.6 except that the nailed wood box shall be class 1, cleated plywood box shall be type IV, class 1, and the triple-wall fiberboard box shall be class 1. The box closure shall be as specified in the appendix of the applicable box specification.

5.2.3 *Level C.* Radio Set AN/ARC-73( ) and individual units shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery of destination. Shipping containers shall comply with the rules or regulations of the common carrier applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping container shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended Use. Radio Set AN/ARC-73 ( ) is intended for use in aircraft for voice communication.

6.2 Ordering Data.

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (Level A, Level B, and Level C).

(d) The specific paragraphs of Section 5, which are applicable to the particular procurement.

(e) Preproduction inspection, preproduction pack(s) as follows:
Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon (including rough handling).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and spare parts required.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/ARC-73W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending decision from the contracting officer on the adequacy of corrective action (see 4.5.4.1.4).

6.5 Location of Operational Inspection. It is desirable that the operational inspection (4.15) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected

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sample units) be sampled and inspected immediately prior to packaging.

6.6 **Definition of Level B.** Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 **Verification Inspection.** Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 **Mounting MT2699( )/ARC-73.** The connector plugs and hardware for the plugs should be packaged and taped to the mounting prior to packaging.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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